

# Geoffrey Lazer

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Portfolio: AI/ML | Available April 2027 | Open to roles across Canada

## Professional Summary

AI/ML and software engineer with an MSc in Computing Science. Builds wildfire and sensor-data pipelines, model experiments and AI backend prototypes with reproducible validation, error analysis and technical documentation.

## Technical Skills

**ML / data:** Python, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SQL, LSTM, U-Net

**AI / backend:** FastAPI, LangGraph, RAG, Pydantic, PostgreSQL, Redis, structured outputs

**Delivery / evaluation:** Git, Docker, GitHub Actions, pytest, MLflow, dbt, model benchmarking

## Professional Experience

### XpertVR

Ontario

*Unity VR / AI Developer - Fixed-term contract*

Oct 2026 – Present

- Current R&D scope: connect AI/LLM communication with live Unity simulation state and translate participant instructions into bounded tasks for VR training.
- Responsibilities include AI guardrails, validation, NPC/speech integration, response-latency optimization and VR performance testing.

### University of Alberta

Edmonton, Alberta

*XR/ML Engineer*

May 2026 – Present

- For ergonomic review, integrated synchronized motion, risk highlighting and spoken guidance into a Unity/OpenXR Quest 3 prototype.
- Integrated authenticated RunPod processing and coaching-package imports to load new participant sessions without rebuilding the APK.
- Implemented model-driven warnings, reproducible study scheduling and CSV logging for construction-safety experiments, delivering signed Quest builds and maintenance guides.

### FireSafe AI

Edmonton, Alberta

*Machine Learning Engineer*

Oct 2025 – May 2026

- Built Python ETL joining Canadian/U.S. wildfire events, weather and geospatial records into standardized ML datasets for wildfire-spread modeling.
- Automated normalization, weather aggregation and event construction, generating consistent 15-day multivariate sequences for training and evaluation.
- Developed multiclass/binary rate-of-spread classifiers and regime-specific regressors, with reusable validation, benchmarking and inference workflows.
- Investigated class imbalance, anomalous predictions and data-quality issues; documented assumptions, failure cases and technical decisions for reproducible team handoff.

### Luxsonic Technologies Inc.

Saskatoon, Saskatchewan

*Software Developer (Internship)*

May 2025 – Aug 2025

- For immersive medical-data review, built Unity/C# volume visualization, OpenXR input and offline Vosk transcription; explored C++/VTK DICOM mesh extraction.

### University of Alberta

Edmonton, Alberta

*Graduate Teaching Assistant, CMPUT 402*

Jan 2025 – Apr 2025

- Guided CMPUT 402 teams in CI/CD and automated testing with GitHub Actions, JUnit, SonarQube and mutation testing, providing actionable software-quality feedback.

### MedWatch Technologies Inc.

Edmonton, Alberta

*AI/ML Researcher*

Oct 2024 – Dec 2024

- For optical-sensor glucose prediction, built sequence preprocessing, LSTM regression and DQN device-configuration experiments, enabling comparison with an LSTM baseline.

## Selected AI & Data Projects

### Agentic Data Assistant for Business Operations

Prototype

- To answer document/database questions, built FastAPI/LangGraph RAG and SQL routing, returning structured answers with sources, tool metadata and trace IDs.
- Implemented SELECT-only SQL, allowlists and row limits to constrain access; added MLflow evaluation, dbt tests, pytest, Docker Compose and CI for repeatable validation.

### Domain-Adaptive Water Body Extraction

Sep 2024 – Dec 2024

- For cross-sensor water extraction, prepared 256x256 Sentinel patches and explored LU-Net/pix2pix adaptation, producing ground-truth mask comparisons for evaluation.

## Education

### University of Alberta

Edmonton, Alberta

MSc, Computing Science

Sep 2024 – Jul 2026

### Loyola-ICAM College of Engineering and Technology

Chennai, India

BE, Computer Science and Engineering

Jul 2020 – Jun 2024